Deep dive on H-T and H-M: carry-up cost decomposition and the Weierstrass-model effect on IC

EXPLORATORY, POST HOC. Written 2026-11-06, after the preregistered analysis and while the independent validation of the frozen snapshot is running. Study `research/volcano_ecdlp_census_followup_20261023/`. Nothing in this document changes a preregistered verdict, a frozen result or `analysis/results.json`, and no statement here is a speedup claim. Numbers come from `results_deepdive.json` (`deepdive.py`, seed 20261106, 10,000 replicates unless stated) over the committed extracts, census files and, for the IC mechanism, the 12 M session archives streamed through `tar`. Unpriced native work is priced at 0, 0.1 and 1 GAE per operation (ANALYSIS_PLAN s1).

Summary (one page)

Preregistered verdicts, unchanged, quoted from `REPORT.md`:

H-T (carry up to the crater, solve with `rho-cubic`): SUPPORTED. Pooled over 63 source vertices (vertex-weighted, A3.2), T = 0.5946 [0.5684, 0.6220] at 0.0 GAE, 0.5917 [0.5655, 0.6193] at 0.1 GAE, 0.5781 [0.5512, 0.6063] at 1.0 GAE per Velu point map. The control T0 (same transport, same crater curve, `rho-neg`) = 0.9672 [0.9227, 1.0145], 0.9672 [0.9223, 1.0149], 0.9670 [0.9204, 1.0164], whose CI includes 1 at all three prices. Theory 1/sqrt(3) = 0.5774; the transport term is 0.21% of the numerator at 1 GAE. Measured same-curve automorphism gain in S (`rho-cubic` / `rho-neg` on the carried instances): 0.6148 [0.5895, 0.6420]. This is a measured end-to-end ratio of a known composition, about 0.58x in ops_gae, not a new speedup; and it is relative to `rho-neg` at the source, which is the strongest rho available on every (j != 0) source curve (erratum F3).

H-M (does IC cost depend on the Weierstrass model?): INCONCLUSIVE by the frozen-seed count, on the Monte Carlo boundary with 'supported for IC' (erratum F1). `ic` is significant (Holm, alpha 0.05) on 5 of 12 chosen curves; `rho-neg` on 0 and `bsgs` on 0. Supported would need 6 of 12 for `ic` and at most 1 for each of the others; refuted would need at most 1 for `ic`. [Withdrawn in the erratum (F1): the sentence 'IC's model dependence is therefore present on some curves and absent on others: it is neither a general property nor noise' is not supported; a precise evaluation of the same test gives 6 of 12, 'supported for IC', so the verdict is not robust either way.] The census's V3 crater values (IC 57 / 67 / 73) are quoted only as the motivating observation.

Erratum F1 of `REPORT.md` (quoted, last sentence of the erratum entry): **The H-M verdict therefore sits on the Monte Carlo boundary between inconclusive and supported for IC and is not robust either way.**

Exploratory findings (all post hoc; n and intervals in the sections):

#	finding	key numbers (n)
1	T factors exactly as T = F x T0; the automorphism factor F carries the gain, the control T0 is consistent with 1	at 1 GAE: F 0.598 [0.571, 0.626], T0 0.967 [0.921, 1.015], T 0.578 [0.551, 0.606]; n = 63 vertices
2	the 6% gap to 1/sqrt3 is a charged additive part (ops minus canonicalisations = 371 +- 8) that does not shrink; the iteration count itself matches theory	charged ratio 0.615 [0.589, 0.642]; canonicalisation ratio 0.578 [0.551, 0.607]; predicted 0.614; n = 3064 pairs
3	per-vertex T_v is homogeneous: no trend with level; 2 of 63 CIs include 1	Cochran Q 65.4 on 62 df, p = 0.36; slope 0.9966 per level, p = 0.74
4	break-even price of a Velu point map is about 177 GAE (charged ops) and has no finite value under the plan's same-price pricing; transport share 0.21% at 1 GAE	model extrapolation (not measured): 4.32e-09 at 64 bits, 5.46e-38 at 256
5	H-T is a known composition: the volcano preserves r, the crater adds the order-6 automorphism for the price of a few point maps	no new method; see section 5
6	significant IC curves (frozen seed): V5:14, V6:2, V6:99, V6:214, V6:223; V6:98 is the sixth	no V5-vs-V6 or level pattern (Fisher p = 0.58); n = 12 curves
7	paired/blocked tests agree with the unpaired ones (blocking gains nothing); pooled across 12 curves the IC model effect is strong, rho-neg and bsgs show none	sum of F 33.6 vs null mean 12.0, p <= 5e-05; rho-neg p = 0.83; n = 6,144 rows per arm
8	max/min of model means is biased up by noise alone (null median 1.13-1.18); excess over null p95 is visible for IC only	IC above null p95 on 9 of 12 curves, rho-neg on 0, bsgs on 0
9	IC cost tracks relation yield, which tracks how many factor-base relation sources fall in the subgroup on each model; the Legendre count (factor-base length) does not	log trials per relation vs log H: r = −0.84; log S vs log H: r = −0.70, p = 0.0001; n = 96 models
10	Q1's curve-effect test and H-M do not contradict: Q1's within-vertex spread already contains the model effect, which Q1's own data show (V6, 648 vertices x 3 models)	pooled F 1.36, permutation p = 0.0005; between-model CV 8.8% [7.2, 10.2]
11	design recommendation: about 96 rows per model (1.5x the 64 used) gives 80% power for >= 6 of 12 under moment-shrunk truth	simulated, 1000 studies per point

1. H-T decomposition (question 1)

Exploratory, post hoc. n = 63 source vertices (levels 1-6 with 1/2/4/8/16/32 vertices), 3,064 (target, round) pairs per arm; vertex-weighted ratios of means as in A3.2; cluster bootstrap, 10,000 replicates, seed 20261106, 95% percentile CI.

With C the vertex-averaged priced `ops_gae`, M the mean number of Velu point maps, p the price per point map and N_v the source rows:

- F = (C(rho-cubic, Tcr) + p M) / (C(rho-neg, Tcr) + p M): the crater factor, the same transport in both terms.
- T0 = (C(rho-neg, Tcr) + p M) / C(rho-neg, N): the control of the plan.
- T = F x T0 holds exactly (maximum error 1.1e-16 at every price, overall and per level).

price per point map	F (crater factor)	T0 (control)	T (verdict quantity)	transport share of T numerator
0 GAE	0.615 [0.589, 0.642]	0.967 [0.923, 1.013]	0.595 [0.569, 0.622]	0.000%
0.1 GAE	0.612 [0.586, 0.639]	0.967 [0.922, 1.013]	0.592 [0.566, 0.619]	0.036%
1 GAE	0.598 [0.571, 0.626]	0.967 [0.921, 1.015]	0.578 [0.551, 0.606]	0.207%

Per level at 1 GAE (descriptive; the plan's verdict is the overall row only):

level	vertices	F	T0	T
1	1	0.609 [0.574, 0.646]	0.956 [0.902, 1.014]	0.582 [0.548, 0.617]
2	2	0.580 [0.546, 0.615]	1.005 [0.944, 1.075]	0.583 [0.547, 0.622]
3	4	0.555 [0.520, 0.594]	1.015 [0.953, 1.086]	0.563 [0.528, 0.600]
4	8	0.608 [0.561, 0.662]	1.007 [0.937, 1.087]	0.612 [0.558, 0.672]
5	16	0.595 [0.553, 0.641]	1.007 [0.917, 1.103]	0.599 [0.549, 0.652]
6	32	0.603 [0.558, 0.653]	0.931 [0.856, 1.010]	0.562 [0.519, 0.608]
all	63	0.598 [0.571, 0.626]	0.967 [0.921, 1.015]	0.578 [0.551, 0.606]

Which factor carries the gain. F does. In logs at 1 GAE, log T = −0.5480 splits into −0.5152 from the pure automorphism ratio (rho-cubic over rho-neg on the crater rows, uncharged work priced), +0.0008 from transport, and −0.0336 from the control: 94% of the log-gain is the automorphism, 6% is T0 sitting below 1. **T0 is consistent with 1**: its CI 0.967 [0.921, 1.015] includes 1 at all three prices, as it should, because the cost of rho depends on the group order r and the volcano preserves r. Its 3% shortfall is consistent with sampling noise: the bootstrap variance of log T is shared between the numerator and the denominator draws (var log F = 5.4e-04, var log T0 = 6.3e-04, covariance −3.0e-04), so the T0 draw does about as much to the width of T's CI as F does. Level 6 alone has T0 = 0.931 (CI upper end 1.010), which pulls its T to 0.562, about level 3's 0.563, the lowest.

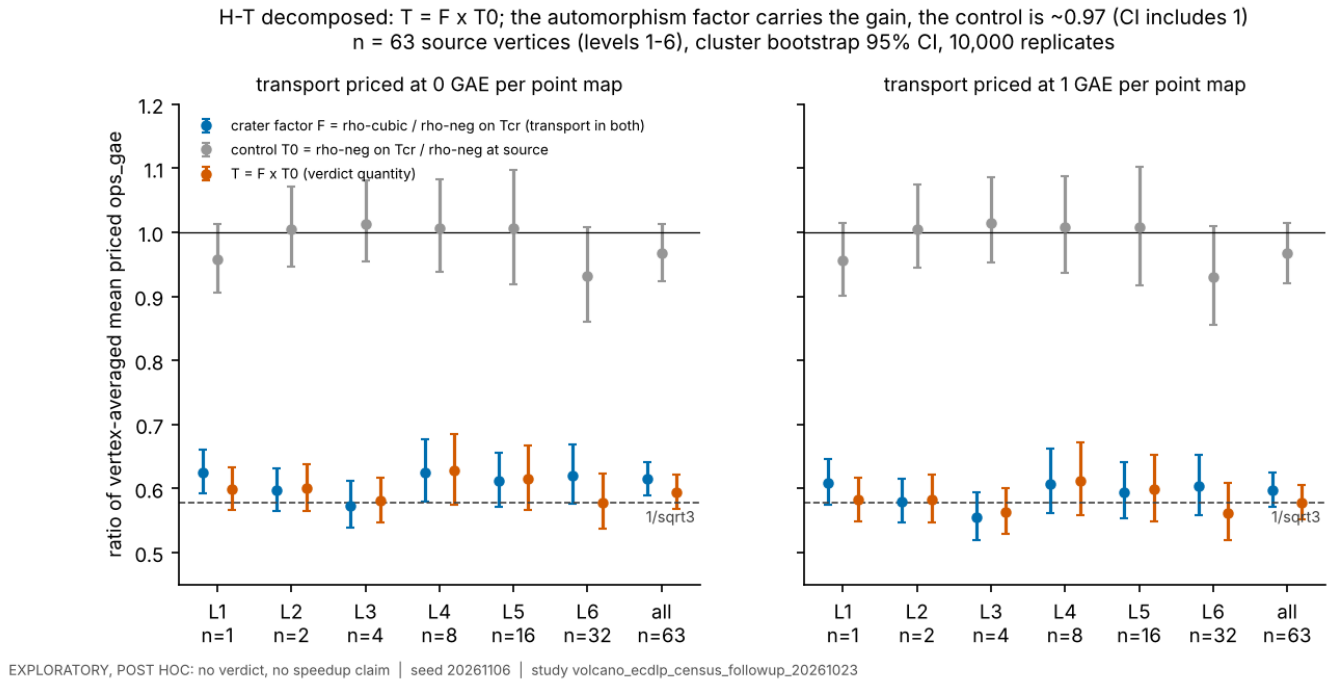


Figure 1: fig1: H-T decomposed (exploratory, post hoc). Source `results_deepdive.json`, key `H-T_decomposition`; SVG beside the PNG.

2. Gap to the 1/sqrt3 theory (question 2)

Exploratory, post hoc. Tcr pairs: n = 3064 (63 vertices); crater positive-control pairs (D, N, ctl pooled): n = 4000. CIs: vertex-cluster bootstrap (Tcr) and pair bootstrap (crater), 10,000 replicates.

Theory: with equivalence classes of order 6 against 2, rho needs sqrt(pi r / 12) against sqrt(pi r / 4) iterations, ratio 1/sqrt3 = 0.5774. Measured charged-ops ratio rho-cubic / rho-neg: **0.615 [0.589, 0.642]** on the carried instances and **0.6113 [0.5992, 0.6236]** on the crater (the report's 0.6113). Both CIs exclude 1/sqrt3: the gap is +0.037 (6.5%) and +0.034 (5.9%).

What explains it, derived from the recorded fields. Every V5 rho row records `ops_gae` (charged) and `canonicalisations_uncharged`. Their difference is:

- 371.1 +- 8.2 operations (min 340, max 394), **identical in the rho-cubic and rho-neg rows of a pair in 100% of the 3064 pairs**, and uncorrelated with the walk length (correlation +0.003 for rho-neg, −0.010 for rho-cubic). It is an additive part that belongs to the instance, not to the walk: the automorphism does not shrink it. (What it is, setup or collision handling, is not derivable from the CSV; only that it does not scale with the walk.)
- The canonicalisation count, the natural proxy for walk iterations, has the ratio **0.578 [0.551, 0.607]** on Tcr and 0.5751 [0.5624, 0.5882] on the crater: both include 1/sqrt3 = 0.5774.
- If the iterations scale by exactly 1/sqrt3 and the additive part stays, the charged ratio would be (theta C + f) / (C + f) = **0.6139** on Tcr (measured 0.6148) and 0.6133 [0.6128, 0.6138] on the crater (measured 0.6113). The gap is therefore accounted for by the additive part, with nothing left over that the data can resolve.

Absolute levels: canonicalisations / sqrt(r) are 0.972 (rho-neg) and 0.555 (rho-cubic) against the idealised sqrt(pi/4) = 0.886 and sqrt(pi/12) = 0.512: both arms sit 9.6% and 8.6% above the formula. That excess (distinguished-point delay, fruitless cycles, collision detection) is equal in the two arms within noise, so it cancels in the ratio; the data **cannot separate** its causes, and in particular cannot say whether rho-cubic loses more to fruitless cycles than rho-neg.

Distribution shape (Tcr pairs, n = 3,064 each):

quantity	mean	median	CV	median/mean	skew
S rho-cubic	0.6464	0.6124	0.449	0.947	0.79
S rho-neg	1.062	1.015	0.457	0.955	0.65
canonicalisations rho-cubic	2268	2130	0.522	0.939	0.79
canonicalisations rho-neg	3966	3775	0.500	0.952	0.65
Rayleigh reference			0.523	0.939	

The canonicalisation counts have the CV and median/mean of a Rayleigh-like collision time in both arms (CV 0.50-0.52 against 0.523), so rho-cubic's walk length is the same shape as rho-neg's, scaled. Quantile ratios cubic / neg (5, 25, 50, 75, 95, 99%): charged S 0.690, 0.616, 0.603, 0.606, 0.597, 0.601; canonicalisations 0.591, 0.555, 0.564, 0.576, 0.577, 0.586. The charged ratio is highest at the low quantiles (the additive part is a larger share of a short solve), the canonicalisation ratio is flat near 0.56-0.59 (row-bootstrap CIs in fig8, n = 1,000 replicates).

Not charged for rho-cubic: the class-representative work per iteration (up to six images) is counted in `canonicalisations_uncharged` for both arms and priced at the same p for both. A canonicalisation under the order-6 action is plausibly dearer than a sign choice; a per-arm price was not measured, so T at 0.1 and 1 GAE treats them alike.

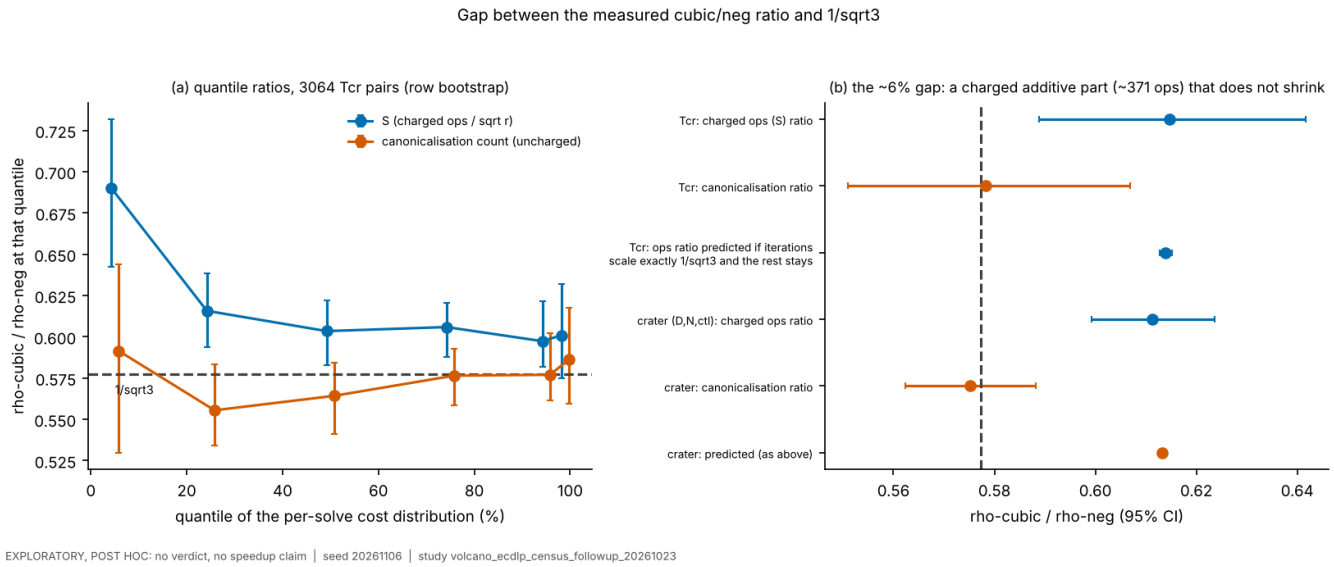


Figure 2: fig8: gap to theory. Source `results_deepdive.json`, keys `H-T_gap`, `crater_rows`.

3. Heterogeneity across the 63 vertices (question 3)

Exploratory, post hoc. n = 63 vertices; per-vertex CIs are the preregistered pair-bootstrap CIs in `analysis/results.json` (descriptive there), reused here.

T_v at 1 GAE ranges 0.43 to 0.89. The per-vertex CI includes 1 for **2 of 63 vertices at 0 GAE, 2 at 0.1 and 2 at 1 GAE** (vertices 50, 57 at 1 GAE; largest upper end 1.22); their T_v are 50: 0.89 [0.65, 1.22], 57: 0.72 [0.46, 1.10] (8 targets x few rounds each). The control T0_v includes 1 for 58 of 63.

Heterogeneity beyond sampling noise is not detected: Cochran's Q on log T_v = 65.4 on 62 df, p = 0.36, I2 = 5% (standard errors from the percentile CI widths, approximate). Trend with source level, which here equals path length to the crater (M_v = 2 x level): weighted slope of log T_v = −0.0034 per level (ratio 0.9966), permutation p = 0.74 (10,000 shuffles); Spearman rho = −0.21, permutation p = 0.10 (2,000 shuffles). Level means of T_v at 1 GAE: L1 0.582, L2 0.583, L3 0.563, L4 0.614, L5 0.603, L6 0.568. No trend, no outlier beyond what 63 noisy ratios give.

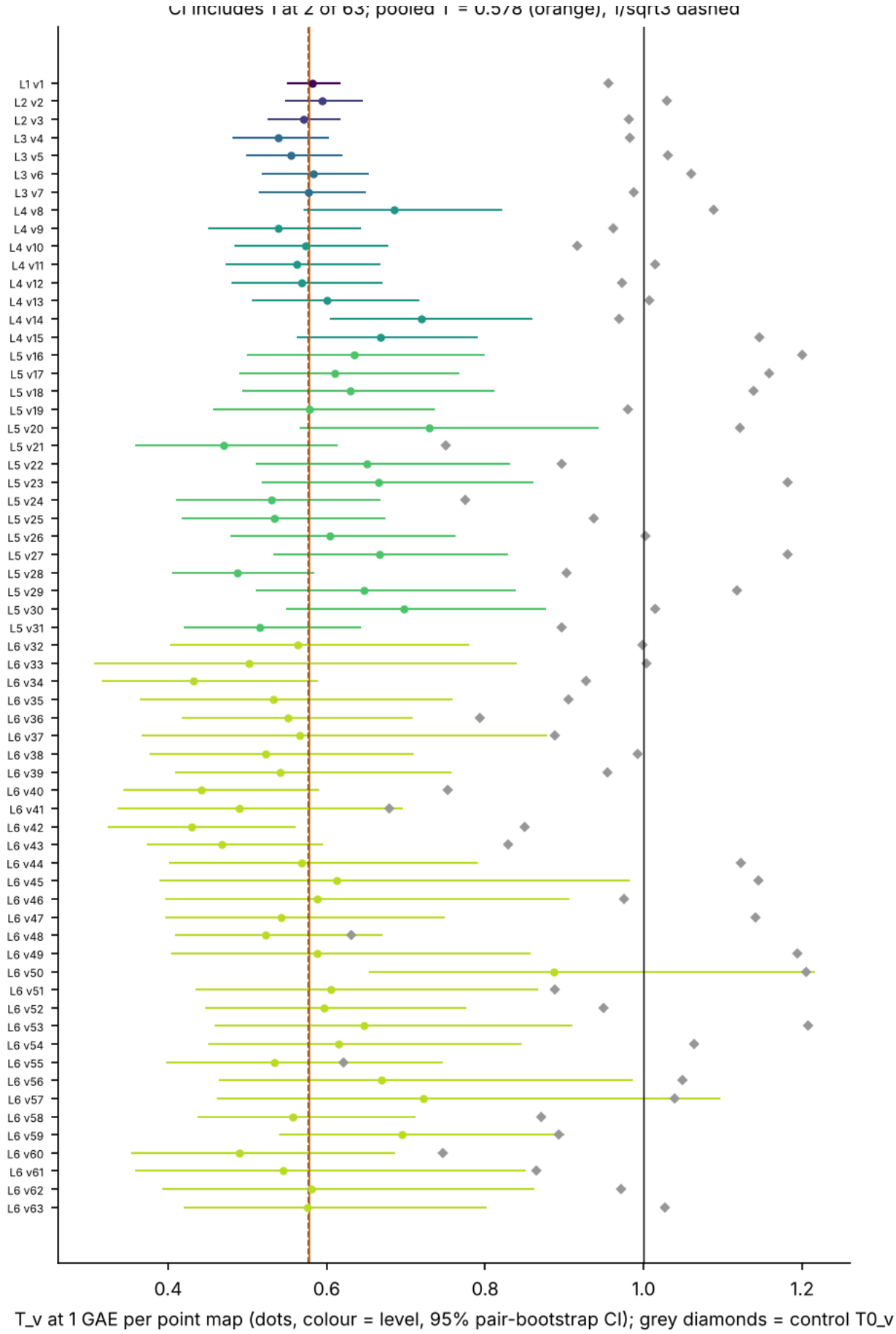


Figure 3: fig2: per-vertex T_v at 1 GAE with 95% CIs, grey diamonds the control T0_v. Source `analysis/results.json` `H-T.by_vertex`.

4. Break-even price and transport share (question 4)

Exploratory, post hoc. n = 63 vertices. Break-even prices are point values per vertex (no CI); pooled transport shares carry the bootstrap CI of section 1.

p* is the price per Velu point map at which T = 1: p* = (C(rho-neg, N_v) - C(rho-cubic, Tcr_v)) / M_v. It depends on what C includes:

- (a) charged ops only: per vertex min 33, median 165, max 901 GAE per point map; **pooled 177 GAE**. By level 1-6: 901, 437, 304, 200, 167, 159.
- (b) every uncharged operation priced at 1 GAE: min 65, median 330, max 1803; pooled 353.
- (c) the plan's definition, the same p on the uncharged counters and on the point maps: **no finite p* exists for any of the 63 vertices**. Raising p adds more to the rho-neg denominator (about 3,900 canonicalisations) than to the rho-cubic numerator (about 2,300) and the transport term (2 to 12), so T falls with p, from 0.5946 at 0 to 0.5781 at 1.

A Velu point map at an order-2 isogeny costs a handful of field operations, so the study's upper price of 1 GAE is already conservative, and the break-even is two to three orders of magnitude above it.

Transport share of the T numerator, measured, at 1 GAE (95% CI): level 1 0.040% [0.038, 0.042]; level 2 0.082% [0.078, 0.086]; level 3 0.128% [0.122, 0.134]; level 4 0.159% [0.149, 0.170]; level 5 0.201% [0.189, 0.213]; level 6 0.246% [0.232, 0.260]; overall 0.207% [0.200, 0.214]. It grows linearly with depth (2 point maps per level) and stays below 0.25%.

MODEL EXTRAPOLATION, not a measurement. Assumptions: solve cost = c sqrt(r) GAE with c = 0.646 (the vertex-averaged rho-cubic S on Tcr; the additive part scales too); transport = 2 x depth point maps priced at p, independent of bit size; depth scenarios: the measured mean depth (5.1), 6 held fixed, and 6 x bits / 26. Not modelled: finding the path to the crater, whether a j = 0 volcano of that depth exists at that size, cofactors, and the class machinery at that size.

bits of r	depth scenario	depth	transport GAE (p = 1)	projected transport share, p = 0.1	p = 1
26 (measured)	measured mean depth	5.1	10	1.92e-04	1.92e-03
26 (measured)	depth 6	6.0	12	2.27e-04	2.26e-03
26 (measured)	depth 6*bits/26	6.0	12	2.27e-04	2.26e-03
64	measured mean depth	5.1	10	3.67e-10	3.67e-09
64	depth 6	6.0	12	4.32e-10	4.32e-09
64	depth 6*bits/26	14.8	30	1.06e-09	1.06e-08
128	measured mean depth	5.1	10	8.55e-20	8.55e-19
128	depth 6	6.0	12	1.01e-19	1.01e-18
128	depth 6*bits/26	29.5	59	4.95e-19	4.95e-18
256	measured mean depth	5.1	10	4.63e-39	4.63e-38
256	depth 6	6.0	12	5.46e-39	5.46e-38
256	depth 6*bits/26	59.1	118	5.37e-38	5.37e-37

The share falls like 2^(-bits/2). The projection says transport stays negligible if the model holds; it does not say a crater of such depth exists at cryptographic size, nor that the path can be found cheaply, and it is not evidence.

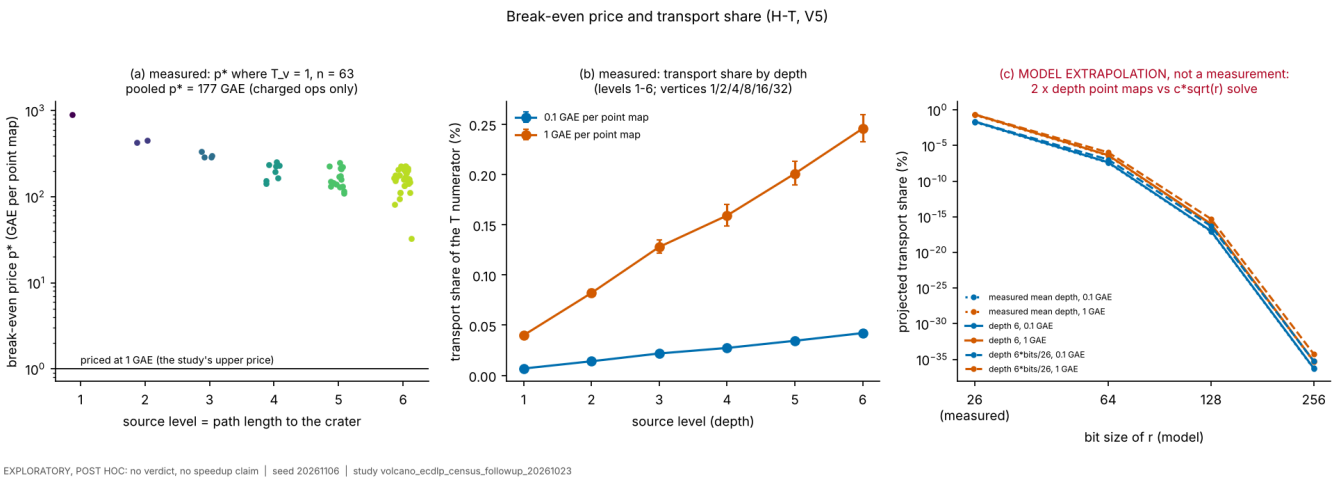


Figure 4: fig3: (a) and (b) measured, (c) model extrapolation. Source `results_deepdive.json`, key `H-T_vertices`.

5. What H-T means (question 5)

An l-isogeny between two curves over the same prime field preserves the group order up to the isogeny's kernel: the volcano's curves all have the same order, so the same prime r in the same size subgroup, and Pollard rho on any of them costs the same, which is what the control T0 = 0.97 (CI includes 1) shows. The j = 0 crater, though, has an extra automorphism of order 6 that the other curves lack; rho can quotient by it and run on classes of size 6 instead of 2, which saves a factor 1/sqrt3, about 0.58. A Velu isogeny is a rational map of known formula, so carrying the instance (G, Q) up to the crater costs two point maps per level: 2 to 12 maps, well under 1% of a 26-bit solve even at 1 GAE each. H-T measures that composition end to end and finds 0.58 of the cost of solving where the instance was born with rho-neg, at 63 vertices.

It is a known composition, the census's A2.3 transport-then-solve with the crater's automorphism as the destination gain, not a new method: the gain is the j = 0 automorphism and the isogeny is only the road to it. It is relative to rho-neg at the source (the strongest rho available on a curve with j not 0 or 1728), and it is a statement about this 26-bit family, where an isogeny path to a j = 0 crater of depth 1 to 6 exists. It says nothing about curves that have no such path, and nothing about index calculus, which is 53 to 141 times the matched rho in this study (REPORT section 9).

6. Which five curves, and where the near misses are (question 6)

Exploratory, post hoc. n = 12 curves, 8 models x 64 (target, round) rows per curve and arm; the table is the frozen-seed result of `analysis/results.json` (permutation p with 10,000 shuffles, seed 20261024, Holm over the 12 curves).

curve	level	ic F	ic p_perm	ic p_Holm (frozen)	significant	rows: unpaired Holm, 20,000 shuffles, new seed	paired Holm
V5:0	0	1.95	0.0629	0.2516	no	0.2372	0.2758
V5:14	4	3.21	0.0014	0.0140	yes	0.0235	0.0215
V5:19	5	1.77	0.0929	0.2787	no	0.2725	0.2791
V5:44	6	0.77	0.6136	0.8623	no	0.8616	0.7867
V6:2	0	3.22	0.0025	0.0225	yes	0.0238	0.0215
V6:43	2	2.21	0.0322	0.1610	no	0.1612	0.2020
V6:98	0	2.81	0.0073	0.0511	no	0.0434	0.0493
V6:99	1	5.23	0.0001	0.0012	yes	0.0006	0.0006
V6:214	1	5.91	0.0001	0.0012	yes	0.0006	0.0006
V6:223	3	3.21	0.0028	0.0225	yes	0.0238	0.0215
V6:262	2	1.00	0.4312	0.8623	no	0.8616	0.7867
V6:599	4	2.33	0.0242	0.1452	no	0.1488	0.1458

The five frozen-seed curves are V5:14 (level 4), V6:2 (0), V6:99 (1), V6:214 (1) and V6:223 (3). Near misses: V6:98 (Holm 0.0511, the sixth), then V6:43 (0.161), V6:599 (0.145), V5:19 (0.279), V5:0 (0.252). **No clustering**: V5 has 1 of 4 curves significant and V6 4 of 8 (Fisher exact p = 0.58); by level (frozen run), level 0 has 1 of 3 (V6:2; V5:0 and V6:98 not), level 1 has 2 of 2, level 2 has 0 of 2, level 3 has 1 of 1, level 4 has 1 of 2 (V5:14 yes, V6:599 no), levels 5 and 6 have 0 of 1 each. Most of the non-significant curves still have F above the null mean of about 1 (V5:0 1.95, V5:19 1.77, V6:43 2.21, V6:599 2.33, V6:98 2.81); only V6:262 (1.00) and V5:44 (0.77) do not. That is why the pooled test of section 7 is the better instrument.

7. Paired and pooled tests (question 7), and the V6:98 boundary

Exploratory, post hoc; seed 20261106. n = 12 curves x 8 models x 64 blocks per arm. The frozen reproduction through the committed `hypothesis_M` matched `results.json` exactly (F and p for all 36 curve-arm cells).

The boundary. The frozen rule reads 5 of 12 as inconclusive; erratum F1 says the 6th curve, V6:98, sits on the Monte Carlo boundary. Here is how far:

- V6:98's raw permutation p is 0.00703 +- 0.00008 at 1,000,000 shuffles (running estimates by shuffle count: 1,000: 0.0060, 10,000: 0.0081, 100,000: 0.0076, 1,000,000: 0.0070); the parametric F(7, 504) gives 0.00699. Holm's multiplier at its rank is 7, so the adjusted p is about 7 x 0.00703 = 0.0492, just under 0.05.
- At 200,000 shuffles per curve the unpaired Holm p is 0.0496 and the unpaired count is 6 of 12; the paired count is 6 of 12 (V6:98 Holm 0.0488).
- 100 independent whole-table reruns with 10,000 shuffles each (the frozen setting, different seeds): unpaired design 6 of 12 in 57 runs and 5 of 12 in 43, V6:98 Holm quantiles 2.5/50/97.5%: 0.0395, 0.0483, 0.0613; paired design 6 of 12 in 71 runs and 5 in 29.

So the preregistered statistic landed on 5 on the one stream it used (43% of independent 10,000-shuffle streams do the same): with an exact evaluation of the same rule the verdict would read 6 of 12, rho-neg 0, bsgs 0, supported for IC. This agrees with erratum F1; it does not change the reported verdict, and it does not rest on any other curve being borderline (at the high-precision setting the sixth curve is at 0.0496 and the seventh, V6:43, at 0.157).

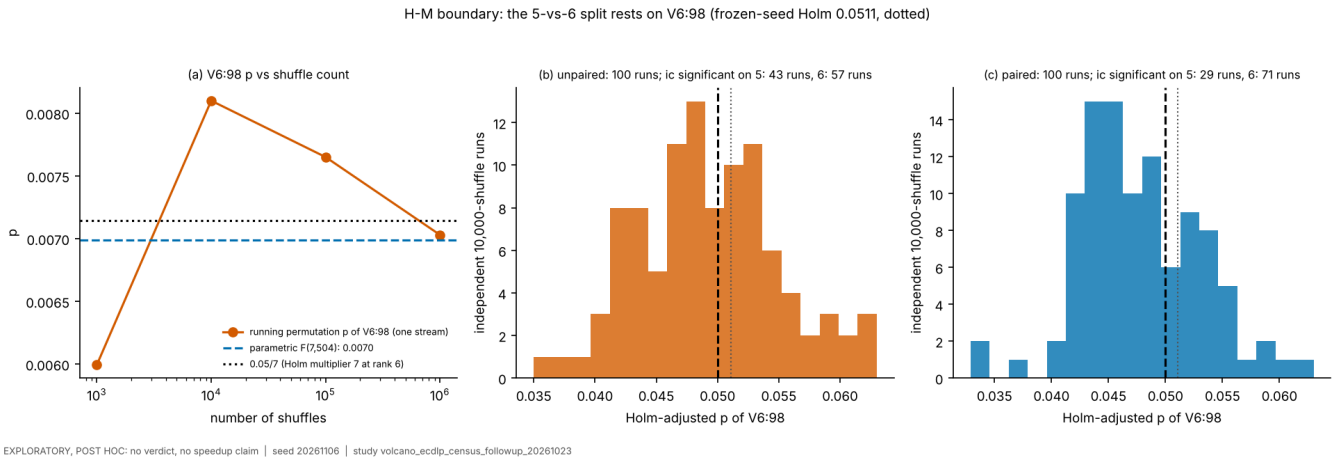


Figure 5: fig7: V6:98 against shuffle count and across independent 10,000-shuffle runs. Source `results_deepdive.json`, `H-M.boundary`.

Paired (blocked) test. The 8 models of a curve solve the same 64 (target, round) instances (BSGS gives identical cost in 100% of blocks), so a blocked permutation (shuffle model labels inside each block) is available. It buys nothing here: the residual mean square after removing blocks is 0.998 of the within-model mean square for IC, and the block share of variance is 0.9% (instance difficulty does not carry across models). Per curve the paired and unpaired p's differ by Monte Carlo and small shifts only (fig4); both designs give **6 of 12** for IC at 20,000 shuffles and 0 for rho-neg; for bsgs every block is identical, so F is 0 (unpaired) and the blocked statistic is undefined.

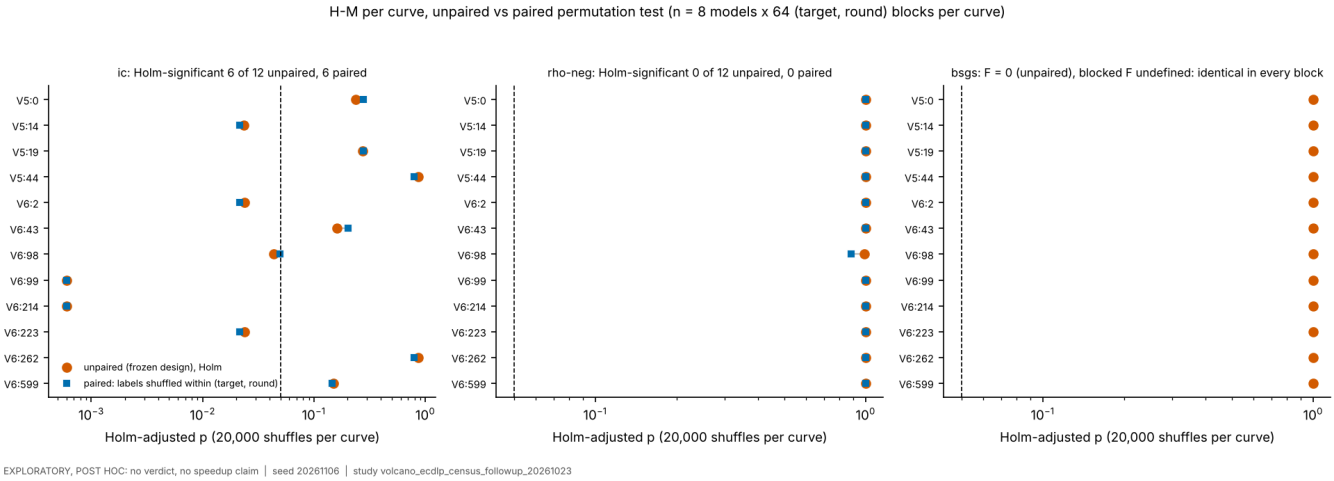


Figure 6: fig4: per-curve Holm p, unpaired against paired, 20,000 shuffles per curve. Source `results_deepdive.json`, `H-M.per_curve`.

Pooled across the 12 curves (the per-curve permutations are independent, so the null is the sum of independent permutation F values; Fisher's combination uses each curve's permutation ranks; 20,000 shuffles, so p cannot go below 5e-5):

arm	design	sum of F (12 curves)	null mean	null 95th pct	p (sum of F)	Fisher p
ic	unpaired	33.6	12.0	15.3	5.0e-05	5.0e-05
ic	paired	33.7	12.1	15.3	5.0e-05	5.0e-05
rho-neg	unpaired	10.2	12.0	15.3	8.3e-01	8.5e-01
rho-neg	paired	10.3	12.0	15.2	8.3e-01	8.4e-01
bsgs	unpaired	0.0	12.0	15.3	1.0e+00	1.0e+00
bsgs	paired	undefined				
ic, excluding the 5 frozen-significant curves (n = 7)	unpaired	12.8	7.0	9.5	4.0e-04	6.0e-04
ic, excluding the 5 frozen-significant curves (n = 7)	paired	12.8	7.0	9.6	4.5e-04	5.5e-04

The evidence for *some* model effect on IC is strong overall (mean F 2.80 per curve against a null mean of about 1.0), and it does not rest on the five significant curves. rho-neg is below its null (mean F 0.85; p = 0.83) and bsgs is exactly 0. The pooled test was chosen after the data; it is a post hoc strengthening of the question, not a replacement for the preregistered per-curve rule.

8. Effect-size calibration of max/min (question 8): a methodological finding

Exploratory, post hoc. n = 12 curves per arm, 8 model means each of 64 rows. Null: model labels permuted across the 512 rows of a curve, 20,000 shuffles, seed 20261106.

`REPORT.md` section 8 gives, per curve and arm, max/min of the eight model means with a percentile bootstrap CI (rows resampled within model). That ratio is biased upward by noise alone: even when no model effect exists, the largest of eight noisy means over the smallest is above 1. Under the permutation null its **median is 1.16 to 1.18 for rho-neg and 1.13 to 1.18 for IC** (95th percentile 1.26 to 1.29). The reported bootstrap CIs inherit the same bias: 12 of 12 rho-neg CIs have a lower end above 1 although rho-neg has no significant curve, and for BSGS, where the models carry identical instances and cost, the observed ratio is exactly 1.000 but all 12 CIs lie above 1 (about [1.03, 1.11]) and 12 of 12 exclude their own point value. The bootstrap CI of a max/min is a CI of a biased statistic, not of the model effect.

arm	observed max/min, range over 12 curves	null median, range	null 95th pct, range	curves above null 95th pct	above null 99th pct
ic	1.118 to 1.577	1.130 to 1.182	1.206 to 1.291	9 of 12	4 of 12
rho-neg	1.099 to 1.259	1.164 to 1.183	1.259 to 1.294	0 of 12	0 of 12
bsgs	1.000 to 1.000	1.064 to 1.066	1.099 to 1.104	0 of 12	0 of 12

Read against the null, **rho-neg's observed 1.10-1.26 is entirely null-sized (0 of 12 above the null 95th percentile)**, so the percentile CIs in REPORT section 8 should not be read as showing a rho-neg model effect, and IC's observed ratio exceeds the null 95th percentile on 9 of 12 curves (4 above the 99th percentile; about 0.6 would be expected by chance): the 'excess over null median' for IC runs from 0.99 (no excess) to 1.35 (V6:214). The unpaired null for bsgs is not a fair benchmark because BSGS values are identical across models by construction (observed 1.000 against a null median 1.06): that cell is an illustration of the bias, not a test. Methodologically, an effect size here should be quoted as observed / null median or as the between-model coefficient of variation (method of moments: IC 7.2% pooled over the 12 curves, rho-neg 1.7%, bsgs 0), not as a raw max/min with a bootstrap CI. I did not edit `REPORT.md`.

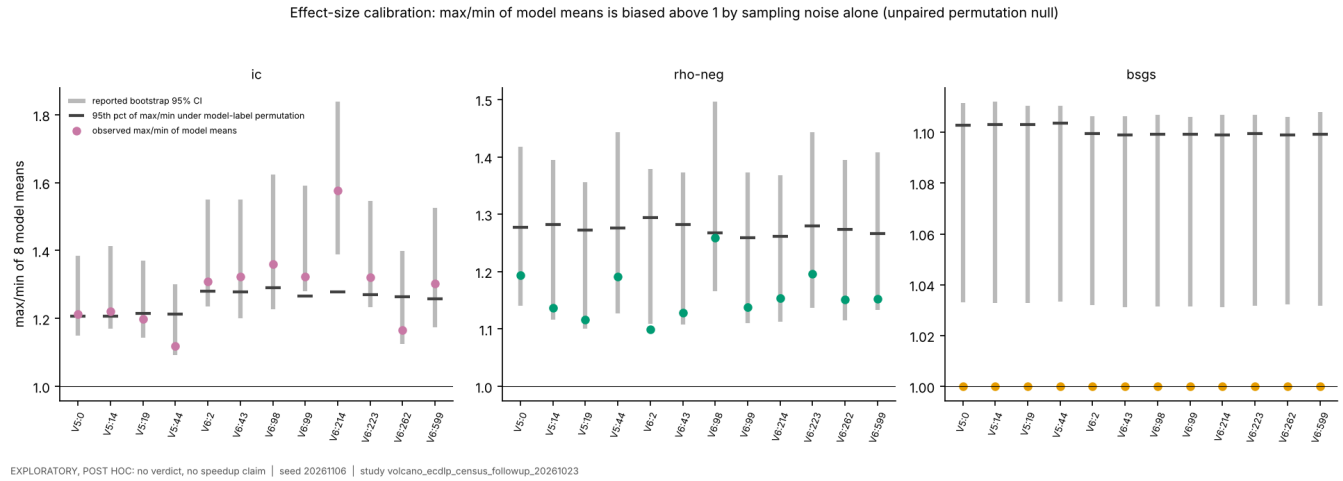


Figure 7: fig5: observed max/min of model means (dots), the reported bootstrap CI (grey) and the 95th percentile under model-label permutation (bars). Source `results_deepdive.json`, `H-M.maxmin_calibration`.

9. Mechanism: why IC depends on the model (question 9)

Exploratory, post hoc. n = 96 models (12 curves x 8), 64 ic rows each. IC features come from the committed extracts; relation counters come from streaming `records.jsonl` of the 12 M session archives (no unpacking). The factor base and the subgroup count H were rebuilt from the census coefficients in 30 lines of exact modular arithmetic (not an ecbench measurement) and checked against the recorded `abscissae_scanned` of all 96 models: all 96 match.

What `prime-abscissa:size=64` is, from `src/cryptanalysis/ic_boundary.rs` (`prime_factor_base`): the 64 smallest abscissae x = 1, 2, 3, ... for which x^3 + ax + b is a square, both signs. 'Prime' means prime field; the base is not prime-valued x. An isomorphism (x, y) -> (u^2 x, u^3 y) moves which small x lie on the curve, so each model has a different factor base. Each IC solve stops after about 19 relations on average and spends almost all its cost in the relation phase: S tracks `trials` (log S against log trials per relation: r = 0.85 within curve).

Findings, within-curve (each curve's models centred, so curve and cofactor effects are removed), permutation p from 10,000 shuffles of model labels inside a curve:

per-model feature	against	r	p
log H (relation sources inside the subgroup)	log trials per relation	−0.841	0.0001
log H	log mean S (ic)	−0.699	0.0001
log H	log mean S (rho-neg), negative control	+0.093	0.3705
k0 + 0.5 (factor-base columns inside the subgroup)	log mean S (ic)	−0.147	0.1005
npair + 0.5 (same-coset column pairs)	log mean S (ic)	−0.090	0.2665
Legendre symbols to build the base (= abscissae scanned)	log mean S (ic)	−0.016	0.89

model coefficient u / p	log mean S (ic)	−0.041	0.71
model coefficient a / p	log mean S (ic)	−0.063	0.58
model coefficient b / p	log mean S (ic)	+0.007	0.95
model coefficient gx / p	log mean S (ic)	−0.004	0.97
model coefficient gy / p	log mean S (ic)	−0.074	0.48
model index 0-7	log mean S (ic)	+0.283	0.009 (one of 13 uncorrected tests; Bonferroni 0.12)

H = 2 k0 + 4 C(k0, 2) + 2 npair counts the distinct elements +-P_i and +-P_i +- P_j (P in the 64 factor-base points) that lie in the subgroup of order r. A relation needs a target combination equal to such an element, so the per-trial yield is expected to be proportional to H / r (a counting argument; the data below test its consequence). H varies by model (for example V5:0 1140-1358, V5:14 1142-1664, V5:19 1188-1578, V5:44 1664-1954), because which small x are on the curve, and in which coset of the subgroup (cofactors 4, 47), changes with the model. Across the 96 models the within-curve correlation between log H and log trials per relation is **−0.84** and with log mean S **−0.70** (within-curve regression slope −0.57, R2 0.49; capped below 1 by row noise on 64-row means). The negative control, rho-neg's cost on the same models, does not correlate (r = +0.09, p = 0.37). The factor-base length (Legendre count) is not the mechanism (r = −0.02), nor are the model coefficients.

Per curve (n = 8 models each; Spearman rank correlation of H with the model's trials per relation and with its mean S; negative means higher H, cheaper IC):

curve	cofactor	rho(H, trials per relation)	rho(H, S ic)
V5:0	4	+0.49	+0.22
V5:14	4	−0.57	−0.57
V5:19	4	−0.86	−0.69
V5:44	4	−0.12	−0.57
V6:2	47	−0.90	−0.86
V6:214	47	−0.98	−0.98
V6:223	47	−0.84	−0.53
V6:262	47	−0.86	−0.71
V6:43	47	−0.90	−0.62
V6:599	47	−0.86	−0.55
V6:98	47	−0.97	−0.87
V6:99	47	−0.91	−0.81

The association is negative on 11 of 12 curves for trials per relation and 11 of 12 for S. It is strongest on V6 (cofactor 47; within-curve r = −0.96 for trials per relation and −0.77 for S, n = 64 models) and weaker on V5 (cofactor 4; −0.36 and −0.47, n = 32; the V5 crater V5:0 has the wrong sign). Neither k0 nor npair alone reaches significance (table above); only their combination H does. The log scale is used so that V5's larger absolute H range does not dominate the pooled within-curve correlation; the rank-based per-curve table shows the pattern without it. That the predictor's form matters is a reason to fix it before testing it again.

This supplies the mechanism that REPORT section 12 declined to claim: the x-coordinate dependence of the factor base acts through how many relation sources fall in the subgroup. It is a measured association on 96 models with a derived predictor chosen after seeing the relation counters, so it is exploratory; a follow-up should fix H before the run. Phase counters other than trials, relations and `abscissae_scanned` (linear algebra rank, row operations) were not used.

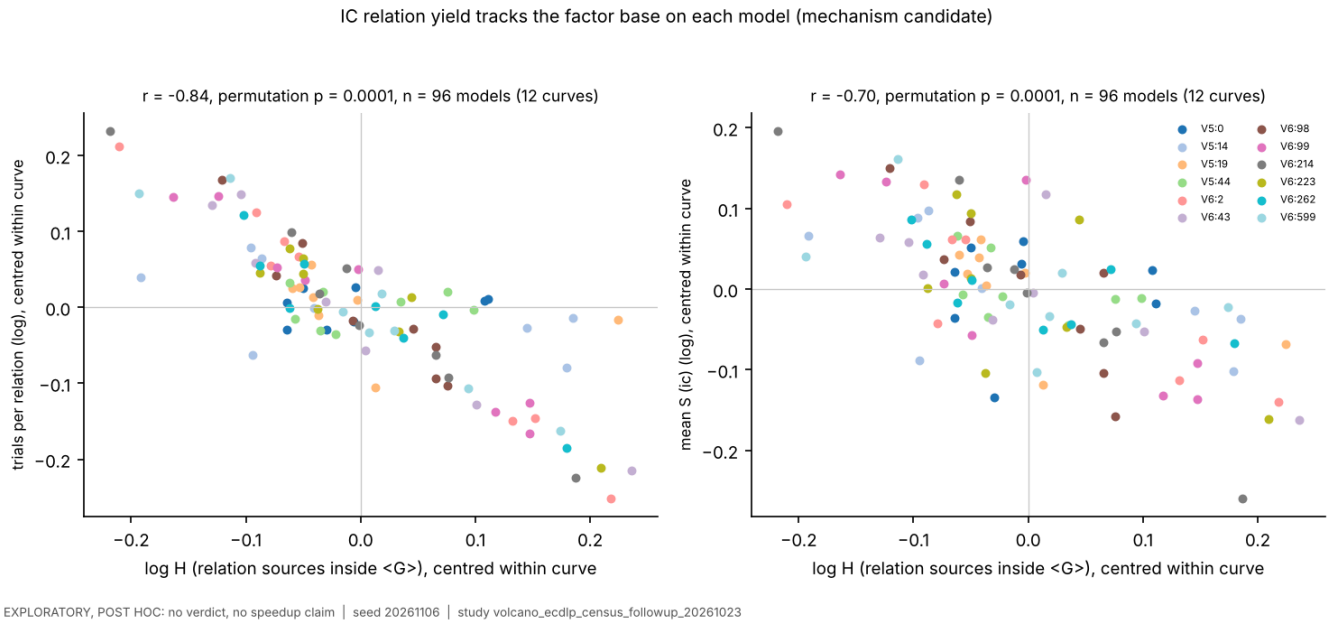


Figure 8: fig11: within-curve log trials per relation (left) and log mean S (right) against log H. Source `results_deepdive.json`, `mechanism`.

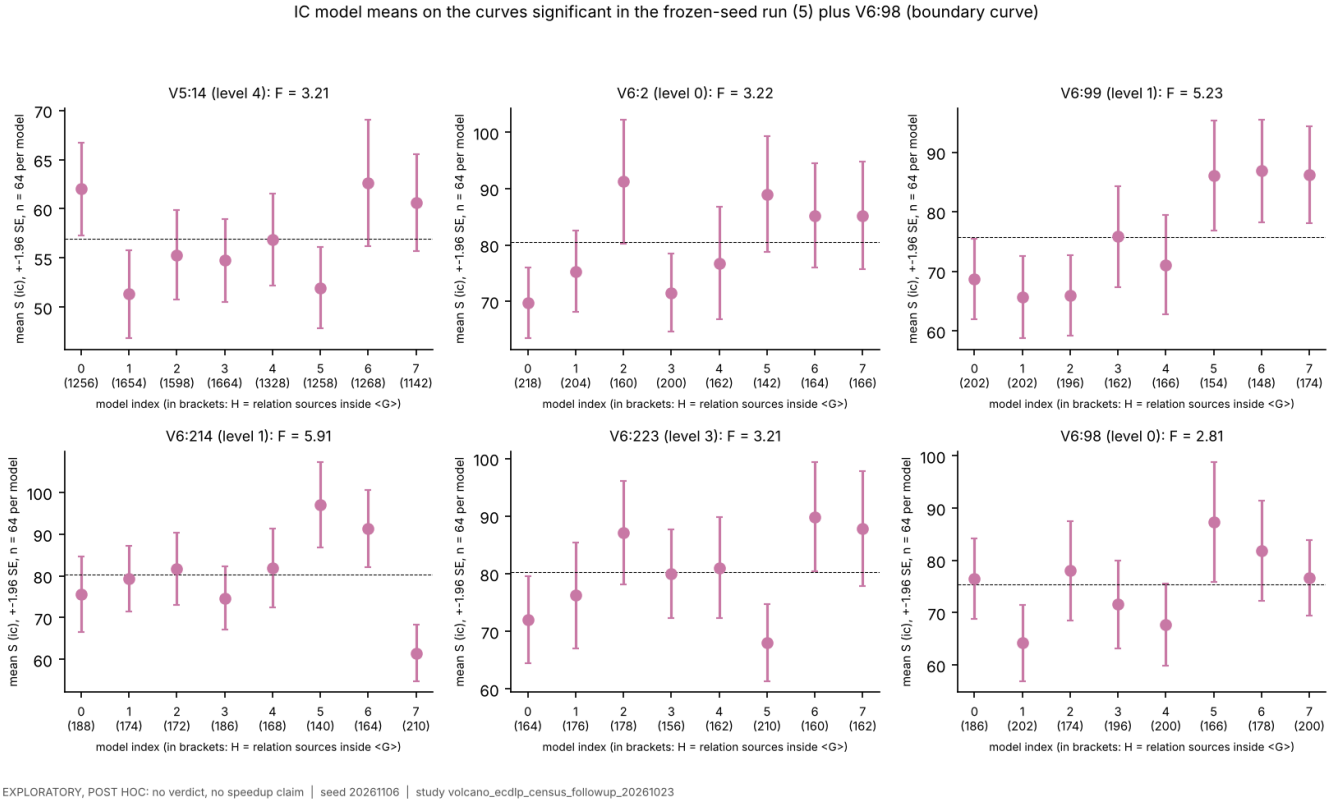


Figure 9: fig6: IC model means (+-1.96 SE, n = 64 rows each) on the five frozen-significant curves and V6:98; brackets under the model index give H.

10. Reconciling with Q1's curve-effect result (question 10)

Exploratory, post hoc. n = 648 V6 vertices, each with its D curve and two ctl re-scalings (three isomorphic models, the same instances); rows per model 8 to 56 depending on level.

The two tests differ in design and in what varies. Q1's curve-effect test asks whether the D curves of different vertices spread more (variance of the D means within level) than the three isomorphic copies of one vertex (within-vertex variance of {D, ctl1, ctl2}); the ratio was F = 0.99 for IC (p 0.42), and erratum F6 notes that its per-curve flags cannot fire at 648 curves (smallest attainable Holm p about 0.33). H-M asks whether the 8 models of one curve differ. They address different questions: Q1 compares *curves* against a yardstick built from *models*, and that yardstick (variance of the three model means, 196.4 for IC) already contains whatever model effect exists. H-M measures the yardstick itself.

Q1's data can be asked the H-M question directly: within each vertex, shuffle the three model labels (unpaired, and inside (target, round) blocks), pooled over vertices:

arm	pooled F (3 models within vertex)	permutation p, unpaired	blocked	between-model CV (95% CI, vertex bootstrap)	within-vertex variance of model means, observed vs row-noise expectation
ic	1.36	0.0005	0.0005	8.8% [7.2, 10.2]	196.4 vs 148.1
rho-neg	0.99	0.6597	0.5997	0.0% [0.0, 3.8]	0.03958 vs 0.04044
bsgs	0 (identical across models)	1.0	1.0	0	0

On Q1's own data the IC model effect is present (p at the 2,000-shuffle floor, 5e-4), with a between-model CV of 8.8% [7.2, 10.2], in line with the 7.2% estimated from the 12 M curves; rho-neg shows none (F 0.99). Of the within-vertex variance of IC model means (196.4), about 148 is row noise and the remaining 48 is model effect; the between-vertex variance of D means was 194.4, which is the same size, so **vertex-to-vertex differences in IC cost are not distinguishable from the model effect plus row noise**. That is what Q1 reports, and it is consistent with H-M: no contradiction. The reading to avoid is 'a curve does not differ from its re-scalings', which suggests the re-scalings are interchangeable; for IC they are not (CV about 9%), and Q1 could not see it because it was built to detect curve effects, not model effects. Q1's rho-neg and bsgs conclusions are unaffected.

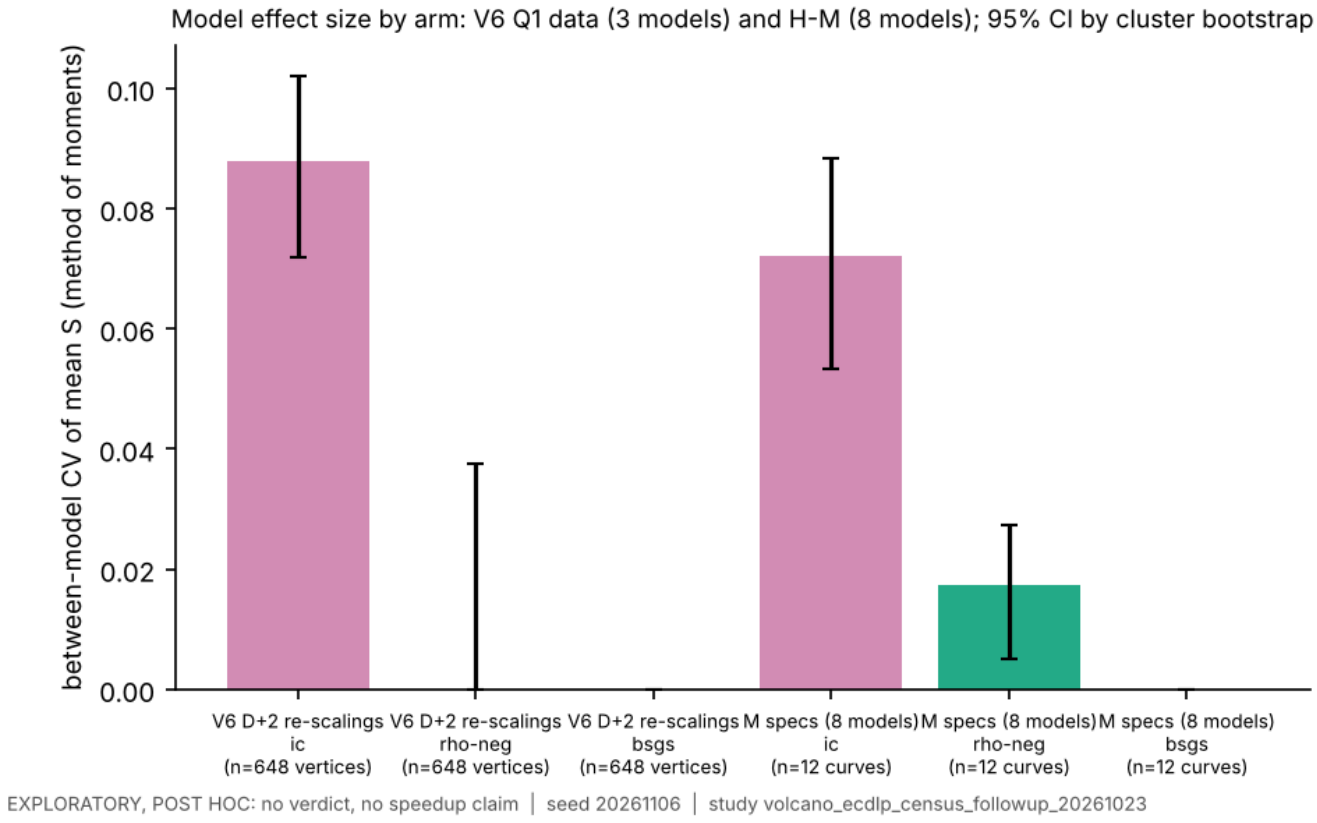


Figure 10: fig10: between-model CV by arm, V6 Q1 data (3 models, 648 vertices) and H-M (8 models, 12 curves). Source `results_deepdive.json`, `V6_model_effect` and `H-M.variance_components`.

11. Power for a follow-up (question 11)

Exploratory, post hoc design recommendation, not a result. 1000 simulated studies per point, seed 20261106.

Simulation: for each of the 12 curves, true model means are the grand mean plus the observed deviations shrunk by sqrt(max(0, 1 - 1/F)) (so that the simulated between-model mean square matches the observed one in expectation), each model's 64 observed rows are rescaled to its true mean and resampled with replacement n times, parametric F(7, 8n-8) p-values, Holm over the 12 curves, outcome = ic significant on at least 6 of 12. 'Observed' uses the unshrunk means (optimistic); 'null' sets all model means equal.

rows per model	P(>= 6 of 12), shrunk	mean significant, shrunk	P(>= 6), observed means	P(>= 6), null
64	0.340	5.0	0.949	0.000
80	0.667	6.2	0.998	0.000
96	0.881	7.1	0.999	0.000
112	0.963	7.9	1.000	0.000
128	0.987	8.4	1.000	0.000

192	1.000	9.5	1.000	0.000
256	1.000	9.9	1.000	0.000
384	1.000	10.1	1.000	0.000
512	1.000	10.0	1.000	0.000

At the 64 rows used, the shrunk scenario has power 0.34 for >= 6 of 12 and expects 5.0 significant curves, which matches the observed 5 or 6: the study sat on the boundary because its power for the 6-of-12 rule was about one third. **96 rows per model** (for example 16 targets x 6 rounds) reaches 80% under the shrunk truth; 64 under the optimistic one. The null scenario never reaches 6 (mean 0.05 false positives of 12), so the rule keeps its size. A follow-up could also fix H (section 9) beforehand and test the yield prediction directly, which needs far fewer rows.

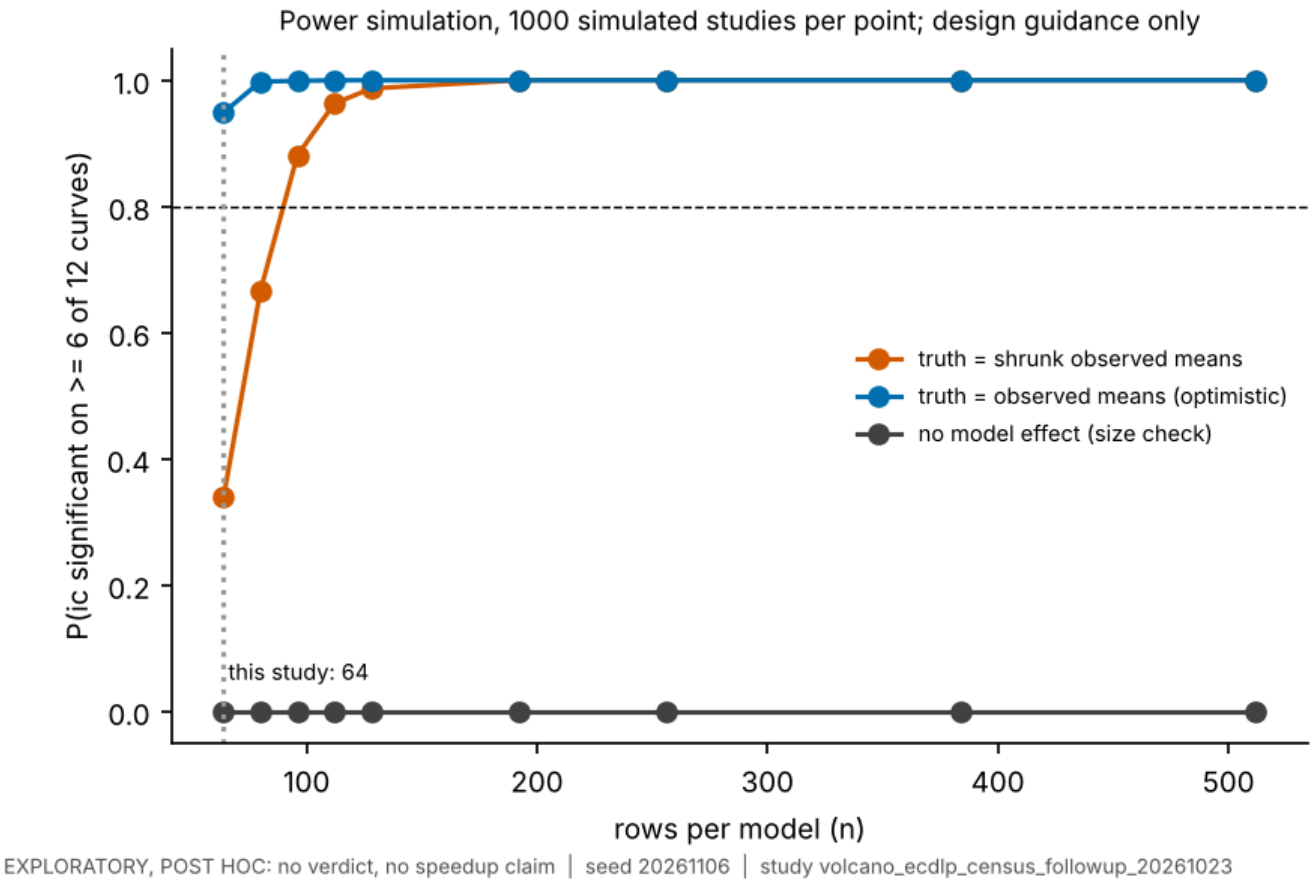


Figure 11: fig9: power of the preregistered H-M rule against rows per model. Source `results_deepdive.json`, `H-M_power`.

Limits

- Everything here is exploratory and post hoc; the analyses were chosen after the frozen results were known, several tests are uncorrected across this document, and no number changes a verdict. The H-M pooled test and the yield mechanism in particular are hypotheses for a follow-up.
- Section 2: the additive part and the canonicalisation proxy are read from recorded counters; their meaning (setup, or something else) is not derivable from them. Canonicalisations are assumed to count walk iterations.
- Section 3: per-vertex SEs for Q come from percentile CI widths and are approximate; 8 targets per vertex.
- Section 4: the extrapolation is a model with stated assumptions; the existence and path-finding cost of deep j = 0 volcanoes at cryptographic size are not modelled.
- Section 7: permutation p-values have Monte Carlo error (stated); the frozen verdict was computed with one stream and is kept.
- Section 9: H is a derived predictor with weights set by counting elements, chosen after the relation counters were seen; the factor base is rebuilt in Python (checked against the 96 recorded scan counts) and is not an ecbench measurement; relation counters come from the M archives only.
- Section 10: Q1's rows per model are small (8 to 56); the pooled F uses all 648 vertices and the 2,000-shuffle permutation has a floor of 5e-4.
- Section 11: the simulation resamples the 64 observed rows per model, so it cannot be heavier-tailed than the data; it assumes the 12 curves are representative.
- One shared VM; validator running concurrently; seeds fixed at 20261106.

Files

`deepdive.py` (reads only committed inputs, writes `results_deepdive.json` with the sha256 of every input), `build_md.py` (lays this file out from the JSON), `build_pdf.sh`, `figures/` (SVG + PNG), `DEEPDIVE.md`, `deepdive.pdf`. Copies of the PDF, SVGs and PNGs are in `benchmarks/20261106-volcano-HT-HM-deepdive/` (no `benchmarks/README.md` row: exploratory, no validation). Run `python3 deepdive.py` (about 8 minutes on 4 vCPUs), then `python3 build_md.py` and `./build_pdf.sh`.